The Triple Illusion Architecture: An Ontological Analysis of Systematic Knowledge Gaps in Large Language Model Interactions

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ABSTRACT

Large Language Models (LLMs) exhibit striking linguistic fluency, which can invite anthropomorphic interpretations and inflated expectations about "understanding." This paper proposes a Triple Illusion Architecture that explains systematic miscalibration in human–LLM interaction across three layers: (Layer I) model-generated illusions arising from a triple approximation (embedding, prediction objective, and reference-frame coverage); (Layer II) human-generated cognitive illusions driven by social attribution and fluency heuristics; and (Layer III) epistemological illusions rooted in the belief that AI has access to complete human knowledge. We formalize a Reference Frame Gap that distinguishes collectible textual knowledge from total human understanding, motivating the *Genius Gap*: LLMs may fail most conspicuously when users operate in novel or underrepresented conceptual frames. We conclude with testable predictions and intervention guidelines that target each illusion layer.

KEYWORDS

Large Language Models, Philosophy of Mind, Chinese Room, Anthropomorphism, Epistemology, AI Safety, Calibration

1 EXECUTIVE SUMMARY AND READER GUIDE

Core claim. Human–LLM interaction produces a *Triple Illusion Architecture*:

- **Layer I(Model-generated).** Illusions arise from (i) embedding approximation, (ii) next-token prediction objectives, and (iii) coverage limits of training data.
- **Layer II(Human-generated).** Humans attribute agency and understanding to fluent text via anthropomorphism and competence heuristics.
- **Layer III(Epistemological).** Users may assume AI can access "all human knowledge," ignoring what cannot be collected, textualized, or represented.

Key novelty. We introduce the Reference Frame Gap and the *Genius Gap*: LLMs can under-serve precisely those users whose conceptual frames lie outside mainstream, collectible corpora.

How to read this paper

Sections ??–?? set the conceptual foundations; Section ?? formalizes Layer I; Section ?? analyzes Layer II–Layer IIIand feedback loops; Section ?? defines an empirical program; Section ?? maps risks and mitigations by layer; Section ?? addresses counterarguments and limitations.

2 SCOPE AND DEFINITIONS

Definition 4 (Collectible vs. total knowledge). Collectible knowledge denotes knowledge that can be captured in accessible artifacts (text, code, diagrams, public records). *Total knowledge* includes tacit, embodied, private, frontier, and culturally under-digitized knowledge not reliably collectible at scale.

Definition 5 (Reference Frame Gap). The Reference Frame Gap is the structural mismatch between the set of reference frames present in training data and the space of reference frames encountered in real users.

UTC/SEF Audit Checklist (for this section)

- Are **"understanding"** and **"truth"** operationalized (what counts as evidence)?
- Are we separating **descriptive claims** (what happens) from **normative claims** (what should be done)?
- Do definitions avoid circularity (e.g., "novel" defined by the model's failure)?

3 THE CHINESE ROOM REVISITED: SYNTAX, SEMANTICS, AND GROUNDING

John Searle's Chinese Room argument distinguishes syntactic manipulation from semantic understanding. Contemporary LLMs can be interpreted as scaled statistical engines that transform text into representations (e.g., vectors) and produce likely continuations, often without grounded access to the external referents language describes.

3.1 What follows (and what does not)

We do *not* claim that fluent behavior is useless or that future systems cannot achieve stronger grounding. We claim a narrower point: *fluent behavior does not guarantee semantic understanding or truth-tracking.*

UTC/SEF Audit Checklist (for this section)

- Is Chinese Room used as an **anchor** (scope-limited) rather than an all-purpose refutation?
- Are alternative views (functionalism, tool-use grounding, embodiment) acknowledged as **counterarguments**?

4 TRIPLE APPROXIMATION MODEL (LAYER I)

We propose that Layer Iillusions can be analyzed as a *triple approximation*:

(1) **Embedding approximation** (statistical proximity ≠ semantic proximity),
(2) **Objective mismatch** (likelihood ≠ truth),
(3) **Coverage mismatch** (collectible ≠ total knowledge; Reference Frame Gap).

4.3 E3: Reference Frame Gap and collectible knowledge limits

Let $D_{\text{collect}} \subset D_{\text{total}}$ be the collectible portion of human knowledge. Reference frames may be systematically missing from D_{collect}, especially for tacit expertise, private frontier insights, or under-digitized cultures.

Hypothesis 1 (Gap sensitivity). As reference-frame novelty increases (measured by a novelty proxy), the probability of ungrounded or frame-drifting responses increases.

Predictions (testable signatures)

- **P1 Reformulation drop:** accuracy declines under semantically equivalent paraphrases.
- **P2 Miscalibration:** confidence does not decline proportionally with accuracy under novelty.
- **P3 Frame drift:** answers regress toward mainstream patterns when prompts encode uncommon ontologies.

UTC/SEF Audit Checklist (for this section)

- Is d_{sem} explicitly operationalized (human judgments / task performance / ontology distance)?
- Are "lower-bound" claims phrased as **hypotheses** rather than theorems unless proven?
- Is "novelty" measurable without circular definitions?

5 LAYER II–LAYER III: HUMAN AND EPISTEMOLOGICAL ILLUSIONS + FEEDBACK LOOPS

5.1 Layer II: Cognitive illusions (human-generated)

Humans may attribute agency and understanding to LLMs via anthropomorphism, fluency heuristics, and authority cues (technical vocabulary, structured prose, confident tone).

5.2 Layer III: Epistemological illusions

Users may assume AI has access to complete human knowledge, overlooking systematic exclusions from training corpora and the existence of tacit and private knowledge.

5.3 Feedback loop

A stylized loop:

$$\begin{aligned}\text{Fluent output} &\longrightarrow \text{Anthropomorphic attribution}\\ &\longrightarrow \text{Overreliance}\\ &\longrightarrow \text{Reinforcement via interface / RLHF.}\end{aligned}$$

Placeholder: causal diagram (DAG) for Layer interactions and feedback.

Figure 1: Causal sketch of Layer I–Layer IIIinteractions and feedback reinforcement.

Code	Failure mode (operational)
F1	Factual hallucination (verifiably false)
F2	Reasoning inconsistency (contradiction across steps)
F3	Citation fabrication (non-existent sources)
F4	Frame mismatch / drift (answers outside user's ontology)

Table 1: Failure mode taxonomy aligned to the Triple Illusion framework.

UTC/SEF Audit Checklist (for this section)

- Do we keep **source-of-error** separate (model vs human vs epistemic assumptions)?
- Are normative recommendations (UI, policy) clearly separated from descriptive psychology?

6 EMPIRICAL PROGRAM: TAXONOMY, TEST SUITES, AND METRICS

6.1 Failure mode taxonomy

6.2 Test suite design

We recommend standardized prompt families:

- **Equivalence under reformulation** (math, logic, definitions),
- **Source-constrained tasks** (history, claims requiring citations),
- **Ontology stress tests** (uncommon premises and concept sets),
- **Abstention quality tests** (when the model should say "I don't know").

6.3 Metrics

- **Accuracy** under verified protocols,
- **Calibration** (confidence vs correctness),
- **Contradiction rate** (self-consistency checks),

- **Frame drift score** (distance from user-stated ontology).

UTC/SEF Audit Checklist (for this section)

- Is every empirical claim tied to a **protocol** and **metric**?
- Do examples serve as **illustrations** rather than proof, unless systematically tested?

7 IMPLICATIONS AND MITIGATIONS (MAPPED TO LAYERS)

7.1 Risks (if–then framing)

- **If** fluency induces overtrust, **then** high-stakes settings face elevated error risk.
- **If** mainstream patterns dominate outputs, **then** intellectual diversity may be pressured toward homogenization (hypothesis to test).
- **If** underrepresented frames are missing, **then** users may experience systematic misrepresentation.

7.2 Mitigation strategies by layer

Layer Iinterventions (model-side / system-side)

- Uncertainty displays and calibrated abstention.
- Source-constrained answering (retrieval + citation verification).
- Consistency checks (self-contradiction detectors).
- Reference-frame novelty detection (novelty proxies + warnings).

Layer IIinterventions (interaction / UI)

- Reduce anthropomorphic framing in interface language.
- Insert lightweight verification prompts in high-risk domains.
- Show counterexamples of known failure modes to calibrate trust.

Layer IIIinterventions (epistemic transparency)

- Disclose training boundary limitations (temporal, cultural, domain).
- Provide "fit-for-purpose" guidance: where the model performs well vs poorly.
- Recommend escalation to human experts for frontier or tacit domains.

UTC/SEF Audit Checklist (for this section)

- Is each mitigation mapped to a specific **edge** in the causal loop (Figure **??**)?
- Are tradeoffs acknowledged (e.g., usability vs caution)?

8 COUNTERARGUMENTS, LIMITATIONS, AND FUTURE WORK

8.1 Counterarguments (brief)

- **Functionalism objection:** if behavior is sufficient, "understanding" may be redefined functionally.
- **Tool-use/grounding objection:** retrieval, action, and perception may reduce Reference Frame Gap effects.
- **Innovation via recombination:** LLMs may still support creativity through novel recombinations of public knowledge.

8.2 Limitations

- The framework is centered on text-only LLMs; multimodal and embodied systems may shift the boundaries.
- Reference Frame Gap measurement requires robust operational proxies (an open methodological task).

8.3 Future work

- Controlled user studies on trust calibration across expertise levels.
- Benchmarks for reference-frame novelty and drift.
- A/B tests for layered mitigations in realistic deployments.

9 CONCLUSION

The Triple Illusion Architecture provides a structured account of why LLMs can be systematically overestimated in interactive settings. By separating model-generated approximations (Layer I) from human attribution mechanisms (Layer II) and epistemological overreach (Layer III), the framework yields testable predictions and targeted interventions. The Reference Frame Gap motivates special caution when models are used to support frontier, tacit, or culturally underrepresented forms of knowing—precisely where the temptation to overtrust fluent output can be strongest.

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